The linear cubic residual has a full q kernel
A covariance bridge to existing embedded ensembles

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Abstract

The first Gaussian-chaos residual of the cubic derivative singlet is a deterministic linear filter of the same couplings as its Hamiltonian. We prove that its joint fixed-degree moments with the Hamiltonian nevertheless tend to those of two orthogonal q-Gaussian variables. The proof approximates the three covariance blocks in rank-weight total variation by finite combinations of existing embedded ensembles of body orders $p, p-1, p-2$. A uniform diagram bound controls the replacement before taking the double-scaled limit. The known Mehler kernel then gives the complete two-point function and the small-coupling limit $\operatorname{sech}^2(2\mathcal{J}_s t)$.

1 Result and place in the singlet construction

Use standard CAR, the complex Gaussian number-preserving p-body Hamiltonian $H = H_s$, all overlapping index sets included, and infinite temperature at zero chemical potential. Time is conjugate to H_s, with $\hbar = 1$. Define

$$T_3 = \tfrac{1}{2}\left(\sum_i c_i^\dagger [H,[H,[H,c_i]]] + \text{h.c.}\right), \qquad F = \mathcal{W}_1 T_3 - \bar{t}(Q_f) H.$$

$\mathcal{W}_1$ is ordinary Gaussian coupling chaos; $\bar{t}$ is the sector-wise ensemble projection coefficient. The preceding notes give its exact rank coefficients, its positive norm and second dynamic moment [1]. Take DS first, $p^2/N = \lambda > 0$, and write

$$q = e^{-4\lambda}, \qquad \sigma^2 = \mathcal{J}_s^2/\lambda, \qquad V = \langle F^2 \rangle_{\text{DS}} = 64 \mathcal{J}_s^6 q^2.$$

For every fixed word w in $h = H/\sigma$ and $y = F/\sqrt{V}$,

$$\boxed{\langle w(h,y) \rangle_{\text{DS}} = \sum_{\substack{\pi \text{ pairs the slots of } w \\ \text{each pair has equal letters}}} q^{\text{cr}(\pi)}.} \tag{1}$$

The sum is zero for odd length. This is a moment statement, not finite-size statistical independence. In particular the full two-point kernel is the existing probe kernel with $r = q$, and

$$\boxed{\lim_{\lambda \to 0^+} \frac{\langle F(t)F(0) \rangle_{\text{DS}}}{V} = \operatorname{sech}^2(2\mathcal{J}_s t).} \tag{2}$$

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2 Rank weights and three covariance blocks

Work first at $m = N/2$, with normalized sector trace. For $r \le \min(m, N-m)$, the unit-variance embedded ensemble has covariance tensor

$$\mathsf{K}_r = \sum_{a=0}^{r} \omega_a^{(r)} \mathsf{Q}_a, \qquad \omega_a^{(r)} = \frac{d_a \binom{m-a}{m-r}\binom{N-r-a}{m-r}}{\binom{N}{m}\binom{m}{r}\binom{N-m+r}{r}}, \tag{3}$$

where $d_a = \binom{N}{a}^2 - \binom{N}{a-1}^2$. Q_a is the covariance of a unit-variance Hermitian Gaussian operator of pure $U(N)$ rank a; in particular $\mathbb{E}A_a^2 = \mathbf{1}$. These are the standard covariance components of the embedded ensemble [2].

Let $C_m = \mathbb{E}\,\mathrm{tr}_m\, H^2$, $H = \sum_a H_a$, $\mathbb{E}\,\mathrm{tr}_m\, H_a^2 = C_m \omega_a^{(p)}$ and $F = \sum_a (t_a - \bar{t}) H_a$. Set

$$c_a = t_a / C_m, \quad \delta_a = c_a - \sum_b \omega_b^{(p)} c_b, \quad W_N = \sum_a \omega_a^{(p)} \delta_a^2.$$

The exactly normalized sector variables are $h_N = H/\sqrt{C_m}$ and $y_N = F/\sqrt{C_m^3 W_N}$. With $u_a = \delta_a/\sqrt{W_N}$, their covariances are

$$\mathsf{K}_{hh} = \sum_a \omega_a \mathsf{Q}_a, \quad \mathsf{K}_{hy} = \sum_a \omega_a u_a \mathsf{Q}_a, \quad \mathsf{K}_{yy} = \sum_a \omega_a u_a^2 \mathsf{Q}_a. \tag{4}$$

Although $\sum \omega_a u_a = 0$, K_{hy} is generally not zero. Consequently scalar energy projection alone does not permit using an independent-probe formula.

The preceding rank calculation proves, with all polynomial moments,

$$s = p - a \Longrightarrow \mathrm{Poisson}(\lambda), \qquad u_{p-s} \longrightarrow \frac{s-\lambda}{\sqrt{\lambda}}. \tag{5}$$

It also supplies uniform bounds $\omega_{p-s} \le K_\lambda (32\lambda)^s / s!$ and $|u_{p-s}| \le K_\lambda (1+s)$ for sufficiently large N.

An exact finite-size ratio connects these polynomial weights to familiar ensembles. For fixed integer $j \ge 0$, define

$$R_j(s) = \frac{\omega_{p-s}^{(p-j)}}{\omega_{p-s}^{(p)}} = \frac{\underline{s^j}\,(N-2p+s+1)^{\overline{j}}\,(N-m+p-j+1)^{\overline{j}}}{(m-p+1)^{\overline{j}}\,(\underline{p^j})^2}. \tag{6}$$

The numerator is zero for $s < j$; bars denote rising and underlines falling factorials. For fixed s, j, $R_j(s) \to \underline{s^j}/\lambda^j$, and $R_j(s) \le K_{\lambda,j}(1+s)^j$ uniformly.

3 The finite covariance bridge

For signed rank covariances define $\|\sum_a z_a \mathsf{Q}_a\|_{\mathrm{rv}} = \sum_a |z_a|$. This measures the total variation of their scalar rank weights. Equations (5)–(6) and the factorial tail give

$$\mathsf{K}_{hh} = \mathsf{K}_p,$$
$$\mathsf{K}_{hy} = \sqrt{\lambda}(\mathsf{K}_{p-1} - \mathsf{K}_p) + o_{\mathrm{rv}}(1), \tag{7}$$
$$\mathsf{K}_{yy} = \lambda \mathsf{K}_{p-2} + (1-2\lambda)\mathsf{K}_{p-1} + \lambda \mathsf{K}_p + o_{\mathrm{rv}}(1). \tag{8}$$

Indeed the associated ratios to $\omega^{(p)}_{p-s}$ tend to $(s-\lambda)/\sqrt{\lambda}$ and $(s-\lambda)^2/\lambda$, respectively. The second equality uses $s^2 = s\underline{2} + s$. Dominated convergence proves the total-variation statements, including the normalization W_N. For $\lambda > 1/2$ one coefficient in (8) is negative. These are signed covariance substitutions in individual Wick diagrams, not a proposed positive Gaussian ensemble.

A bound for an entire diagram. Consider a fixed pairing π with n chords. For positive unit-variance covariance on each chord, represent distinct chords by independent Hermitian Gaussian matrices $A_1, \dots, A_n$. The diagram is the expectation of a normalized trace word in which each A_j occurs twice. Tracial Hölder and the Gaussian matrix bound $\mathbb{E}\,\mathrm{tr}\, A_j^{2n} \le (2n-1)!!$ give

$$|D_\pi(\mathsf{Q}_{a_1}, \dots, \mathsf{Q}_{a_n})| \le (2n-1)!! =: B_n. \tag{9}$$

The Gaussian bound was proved by integration by parts in the preceding dynamics note; it uses only $\mathbb{E}A_j^2 = \mathbf{1}$. Expanding signed rank weights extends (9) to $B_n \prod_j \|\mathsf{K}_j\|_{\mathrm{rv}}$.

Let ϵ_N be the larger error in (7), (8). The actual covariance norms are at most one; the substituted ones are at most

$$B = \max\{1, 2\sqrt{\lambda}, 2\lambda + |1-2\lambda|\}.$$

A telescoping replacement of the n edges therefore changes one diagram by at most

$$n B_n B^{n-1} \epsilon_N \longrightarrow 0. \tag{10}$$

This bound is uniform in the ranks being summed. Thus the covariance bridge controls arbitrary fixed words, rather than only one crossing or a finite list of frequency moments.

The imported diagram law. For independent unit-variance embedded ensembles with body orders $r_j = p - j_j$, where every j_j is fixed, the existing DS chord rule gives $D_\pi \to q^{\mathrm{cr}(\pi)}$. In the general-body rule the pair crossing weight is $\exp[-r_i r_j/(N\rho(1-\rho))]$, $\rho = m/N$; here $\rho = 1/2$ and $r_i/p \to 1$ [3, 4]. The number of chords is held fixed. The covariance bridge reduces our correlated operator to precisely these established independent-ensemble diagrams.

4 All-order cancellation and a rank-filter interpretation

After (7)–(8), each diagram is a finite sum of body-shift choices. Every choice has the same limiting factor $q^{\mathrm{cr}(\pi)}$. The coefficient sum on one edge is

$$hh : 1, \qquad hy : \sqrt{\lambda}(1-1) = 0, \qquad yy : \lambda + (1-2\lambda) + \lambda = 1. \tag{11}$$

Consequently any pairing containing a mixed h–y edge vanishes in the limit. The other pairings each give the usual q^{cr}. Together with (10), this proves (1). It does not assume that Gaussian chaos is preserved under time evolution.

The full infinite-temperature trace follows by restricting first to $|m - N/2| \le N^{2/3}$. On that window the covariance limits, sector normalizations and diagram bounds are uniform. Outside it, the binomial charge probability decays faster than the polynomial bounds on t_a and all fixed Gaussian moments. Thus the full-trace h, y in (1) have the stated limit.

For $P_{uv}(q, r)$, the standard unit-variance probe moment, the immediate two-insertion consequence is

$$\langle F H^u F H^v \rangle_{\mathrm{DS}} = V \sigma^{u+v} P_{uv}(q, q). \tag{12}$$

At total Hamiltonian degree four,

$$P_{40} = 2 + q, \qquad P_{31} = q(2+q), \qquad P_{22} = 1 + q^2(1+q), \tag{13}$$

$$\mu_{4,F} = \frac{2\mathcal{J}_s^4}{\lambda^2}(1-q)^2(3q+5) \longrightarrow 256\mathcal{J}_s^4. \tag{14}$$

Also $\langle y^4 \rangle = 2 + q$ and $\langle hyhy \rangle = q$.

A useful extension of the construction. For fixed polynomials P_i, consider the rank filters $Y_{i,N} = \sum_a P_i(p-a)H_a/\sqrt{C_m}$. Each edge covariance contains $P_i(s)P_j(s)$. Expand this finite polynomial in falling factorials and use (6) with the required finite set of shifts. The same diagram argument gives a q-Gaussian family with covariance

$$G_{ij} = \mathbb{E}_{s\sim\mathrm{Poisson}(\lambda)} P_i(s) P_j(s).$$

In particular orthonormal Charlier polynomials produce orthogonal variables in this limiting moment algebra. h corresponds to $P_0 = 1$, and y to $P_1 = (s-\lambda)/\sqrt{\lambda}$. This extension concerns explicit rank filters of H. Identifying which of them arise in the local derivative-singlet tower is a separate physical construction, not an automatic particle count.

5 The existing kernel now applies to F

Set $(z;q)_\infty = \prod_{j\ge 0}(1 - zq^j)$, and define

$$\rho_q(\theta) = \frac{(q, e^{2i\theta}, e^{-2i\theta}; q)_\infty}{2\pi}, \qquad E(\theta) = \frac{2\sigma\cos\theta}{\sqrt{1-q}}, \tag{15}$$

$$\mathcal{K}_r(\theta,\phi) = \frac{(r^2;q)_\infty}{\prod_{\varepsilon,\varepsilon'=\pm1}(re^{i(\varepsilon\theta+\varepsilon'\phi)};q)_\infty}, \qquad r = q. \tag{16}$$

The known positive Mehler measure [3, 4] gives

$$C_F(t) = \int_0^\pi\!\!\int_0^\pi \rho_q(\theta)\rho_q(\phi)\mathcal{K}_q(\theta,\phi)e^{it(E(\theta)-E(\phi))}\,d\theta\,d\phi. \tag{17}$$

The normalization is $C_F(0) = 1$. This formula is imported after proving the operator-to-probe correspondence, not asserted from the second frequency moment.

At finite size the normalized two-energy measure is $V_N^{-1}\mathbb{E}\,\mathrm{tr}[F\,d\Pi_H(E)F\,d\Pi_H(E')]$. It is positive. Equation (12) determines all its limiting mixed moments. The second energy moments give tightness; higher even moments give uniform integrability for each monomial. The candidate measure in (17) has compact support and is moment determinate, so the measures converge weakly to it. The bounded continuous characteristic function proves (17) at every fixed real time.

For $0 < q < 1$ this measure is absolutely continuous. Its frequency pushforward has support $|\omega| \le 4\sigma/\sqrt{1-q}$ and no atom at zero. Portmanteau on the closed diagonal shows that the normalized finite-size exactly static weight tends to zero as well. This strengthens the previous result, which only established positive surviving dynamic weight.

For stable evaluation the equivalent Jacobi representation is

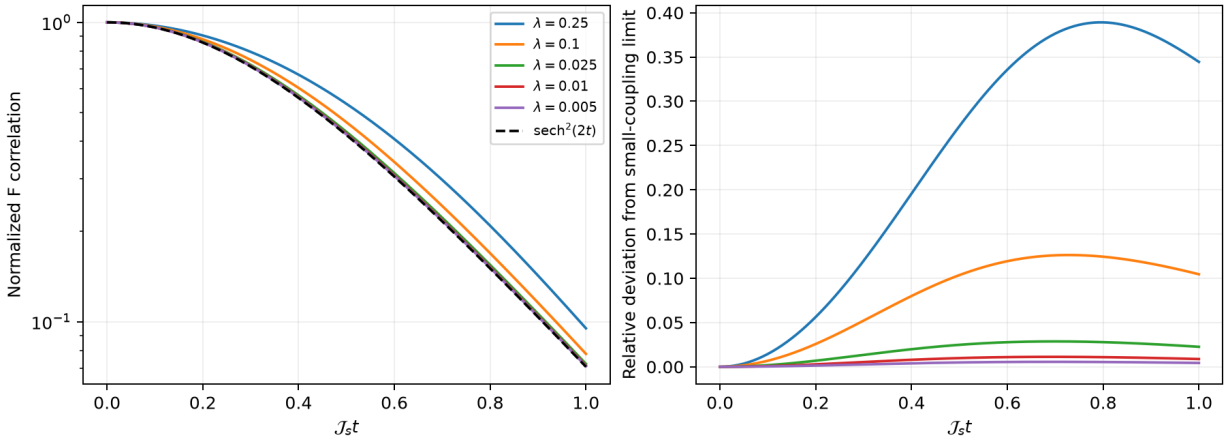
$$C_F(t) = \sum_{n\ge 0} q^n |\langle n|e^{-itT_q}|0\rangle|^2, \qquad (T_q)_{n-1,n} = \sigma\sqrt{\frac{1-q^n}{1-q}}. \tag{18}$$

The known small-coupling probe limit with $r = q^d$, or the fixed-descendant contraction in our preceding note, applies with $d = 1$ [5, 1]. Its Liouvillian ladder entries are $2\mathcal{J}_s\sqrt{(n+1)(n+2)}$, giving (2). In the convention $C(t) = \int S(\omega)e^{it\omega}d\omega/(2\pi)$,

$$S_1(\omega) = \frac{\pi\omega}{4\mathcal{J}_s^2\sinh(\pi\omega/(4\mathcal{J}_s))}, \qquad S_1(0) = \frac{1}{\mathcal{J}_s}. \tag{19}$$

This is a frequency spectrum in string-time units, not a bulk mass spectrum. The equal-time ratio of the F and R strengths remains $2\lambda + o(\lambda)$, as obtained previously.

6 Computed evidence and next physical calculation



The full-rank covariance replacement errors are evaluated without a deficit cutoff. At $p = 256$:

λ	$\|\Delta \mathsf{K}_{hy}\|_{\mathrm{rv}}$	$\|\Delta \mathsf{K}_{yy}\|_{\mathrm{rv}}$
0.10	$8.4170\,10^{-5}$	$9.3549\,10^{-4}$
0.25	$6.5917\,10^{-4}$	$1.2584\,10^{-3}$
0.50	$2.4968\,10^{-3}$	$3.8296\,10^{-3}$
1.00	$8.5851\,10^{-3}$	$1.7252\,10^{-2}$

There are 2,286 exact rational ratio checks of (6) and 20 full-rank cases at 80/120 working digits. Agreement is checked on 65-digit saved values; it is not a rounding certificate.

The finite-CAR calculation evaluates every one of the 15 Gaussian pairings in each six-slot word FH^uFH^{4-u}, in three systems. It retains the mixed-pair contributions separately and rechecks the preceding degree-two results. For $(N, p, m) = (5, 2, 3)$ and coupling variance one, it gives the exact values $\langle F^2 \rangle = 180$ and $\langle FL^4F \rangle = 314280$. These finite-size numbers are not equated to their DS limits.

Five Jacobi computations use $\lambda = .25, .1, .025, .01, .005$, cutoffs $33, 52, 138, 302, 574$, and refinements by 16 levels. The analytic Taylor-tail bound is below 10^{-13}. Independent 128/192-node Mehler quadrature verifies the function and its first two nontrivial moments for $\lambda = .25, .1$. Four Fourier integrals are checked at 50/80 digits. At $\lambda = .005$ the largest relative difference from $\mathrm{sech}^2(2t)$ on $0 \le t \le 1$ is still about 0.560%. All finite-coupling differences are retained. Float refinement and quadrature comparisons are empirical precision checks.

The next calculation is the real-time cross spectrum of F with the cubic third-chaos terms R, S, Z. The theorem here concerns the (H, F) moment algebra; it does not yet combine the raw $M_3 = R + aS + bZ + \bar{t}(Q_f)H + F + \frac{3}{2}LA$ spectrum. After that, the mixed X, R, F interactions and the spatial translation dictionary remain the route to the Susskind bulk construction. This stage was saved locally without a Git commit or push.

Sources and proof dependencies

The proof imports the standard independent-ensemble chord law, not finite-size independence of F. The additional step is the rank-weight covariance approximation and its diagram bound. The asymptotic rank distribution and the exact filter t_a come from our preceding local notes. The Mehler formula and small-coupling probe kernel are existing results.

Nine further full-rank computations evaluate the entire quartic polynomial $\langle (h_N + z y_N)^4 \rangle$, including its odd mixed coefficients, at 80/120 digits. They approach $(2+q)(1+z^2)^2$; all finite-size deviations are stored. This directly tests the joint-law prediction in addition to the two-insertion curvature.

The first curve-generation run stopped during plotting because of a malformed math-text axis label. The label was corrected; the numerical tolerances were unchanged. The failure record, successful data, code hashes, compiler logs and reviewed page hashes are retained with the local evidence.

References

[1] A. Day and Chengheng, local notes: *The missing linear component of the cubic singlet*; *Dynamics of the linear cubic residual*; *A positive chord generator from the tracial seed*. See `manuscript/susskind-cubic-linear`, `susskind-cubic-dynamics`, and `susskind-antisymmetric`.

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[3] J. Tall and S. Tomsovic, *Double-scaled bosonic and fermionic embedded ensembles, complex SYK, and the dual Hilbert space*, arXiv:2604.14522v2, Sec. 2.1, Eqs. (2.17)–(2.21); Sec. 4.1, Eqs. (4.3), (4.15)–(4.18).

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